

Comparative Trial Report

Erythropoietin (ESA) Benefit in CKD-Related Anaemia:

A Three-Query Comparison of Plain LLM Output, Module 45 (Evidence-Based Medicine Insights), and Module 45 + Shadow Module 45 (Adversarial Counterpart)

1. Background and Trial Design

This report compares three responses to functionally the same clinical question — the evidence for erythropoietin (ESA) benefit in CKD-related anaemia — generated under three different querying conditions, treated here as three arms of a single comparison trial.

- Query 1: “Give evidence for erythropoietin benefits in CKD anaemic patients” — answered as a plain, unscaffolded clinical summary.
- Query 2: The same question, run through Module 45 — Evidence-Based Medicine Insights (avi33tbtt.github.io/Prompts/Module-45-Evidence-Based-Medicine-Insights.html), which executes a structured EBM consult cycle (PICO, evidence hierarchy, risk-of-bias appraisal, GRADE certainty rating, three-circle integration, and an insight brief).
- Query 3: The same question, run through Module 45 together with its Shadow Module 45 — Adversarial Counterpart (avi33tbtt.github.io/Prompts/Shadow-Module-45-EBM-Adversarial-Counterpart.html), which layers an adversarial challenge onto each step of the Module 45 consult and produces a ranked Shadow Inventory of findings.

The aim is to evaluate whether — and how — adding structured EBM scaffolding, and then an adversarial critique layer on top of that scaffolding, changes the accuracy, rigor, and clinical usability of the output, using the same underlying evidence base (CHOIR, CREATE, TREAT, Besarab 1998 / NHCT, and subsequent meta-analyses) across all three arms.

2. Summary of Each Arm

2.1 Query 1 — Plain Output

Presents the established benefits of ESA therapy (anaemia correction, transfusion avoidance, quality-of-life gains in CREATE, and the iron-co-therapy finding from PIVOTAL) alongside the cautionary landmark-trial evidence (NHCT, CHOIR, CREATE, TREAT) in a single narrative pass, with a summary table of the four major RCTs and a closing reference to the KDIGO 2026 guideline position. No formal appraisal framework, certainty grading, or bias assessment is applied; conclusions are stated directly.

2.2 Query 2 — Module 45 (EBM Insights)

Runs a full structured consult: PICO definition, evidence hierarchy placement (CEBM 1a/1b), risk-of-bias appraisal by domain (RoB 2 framework), GRADE certainty rating by outcome, an effect-size translation into absolute and relative terms, three-circle integration (evidence, clinician judgement, patient values), and a closing EBM Insight Brief. Grades the mortality/CV harm signal at higher Hb targets as Moderate certainty, citing inconsistency between TREAT

and CHOIR/CREATE on stroke risk as the main downgrading factor. Explicitly flags that key numbers should be checked against primary trial publications before use in teaching material.

2.3 Query 3 — Module 45 + Shadow Module 45

Runs the same structured consult, with the Shadow module challenging each step. Opens by naming a framing trap: that “does ESA correct haemoglobin / avoid transfusion” (yes) and “does ESA improve mortality, cardiovascular outcomes, or quality of life” (no, and trending toward harm at higher targets) are different questions that a single “benefit” answer can wrongly collapse into one. Pulls in an additional 9-trial dose-response meta-analysis (n=5143) showing higher ESA dose independently associated with mortality and cardiovascular harm, and on that basis grades the same mortality/CV harm signal as High certainty rather than Moderate. Closes with a ranked Shadow Inventory of held, moderately-held, and partially-held findings.

3. Evidence Base Referenced Across All Three Arms

The same four landmark RCTs anchor every arm of this comparison, regardless of querying method:

Trial	Population	Comparison	Key finding
NHCT (1998)	Hemodialysis	Hct 42% vs 30%	Trend toward higher mortality and non-fatal MI in the normal-hematocrit group; trial halted early
CHOIR (2006)	Non-dialysis CKD	Hb 13.5 vs 11.3 g/dL	Terminated early; more death and CV events in the higher-Hb group (HR 1.34, 95% CI 1.03-1.74, p=0.03)
CREATE (2006)	Non-dialysis CKD	Hb 13-15 vs 10.5-11.5 g/dL	No difference in primary CV endpoint, but more CV events and dialysis initiations in the higher-Hb arm
TREAT (2009)	Non-dialysis diabetic CKD	Darbepoetin (target Hb 13) vs placebo	No mortality difference; stroke risk nearly doubled (HR 1.92, 95% CI 1.38-2.68, p<0.001)

All three arms also note a consistent secondary benefit: a higher-dose, proactive intravenous iron strategy (PIVOTAL trial) improved cardiovascular outcomes and mortality compared with a lower-dose reactive strategy in haemodialysis patients already on ESAs — underscoring that ESA performance is iron-replete-dependent.

4. Structural Comparison

	Query 1 (Plain)	Query 2 (Module 45)	Query 3 (Module 45 + Shadow)
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Framework used	None — narrative summary	Full EBM consult cycle (PICO -> hierarchy -> RoB2 -> GRADE -> 3-circle -> brief)	Same as Query 2, plus an adversarial Shadow challenge layered onto every step
Trials cited	NHCT, CHOIR, CREATE, TREAT, PIVOTAL (iron)	Besarab 1998, CHOIR, CREATE, TREAT, Cochrane 2023 network meta-analysis	Same four trials, Cochrane 2023 NMA, plus a separate 9-trial dose-response meta-analysis (n=5143)
Recent guideline context	KDIGO 2026 update referenced; HIF-PHIs flagged as emerging alternative	Not referenced	Not referenced
Self-audit of own numbers	None	Explicit flag that figures need primary-source verification	Explicit flag, plus a ranked Shadow Inventory of held/partially-held findings

5. The Key Divergence: GRADE Certainty Disagreement

The most consequential finding from this trial is not where the three arms agree, but where Query 2 and Query 3 disagree despite drawing on largely the same trial base and running the same underlying module.

Outcome domain	Query 2 (Module 45)	Query 3 (Module 45 + Shadow)
Mortality/CV harm at higher Hb targets	Moderate — downgraded for inconsistency (TREAT vs. CHOIR/CREATE on stroke risk)	High — upgraded on a consistent dose-response relationship across four RCTs and meta-analyses
Transfusion avoidance	High	High
Quality of life	Low-to-Moderate	Low-to-Moderate, flagged as contested in a dedicated QoL systematic review
Mortality benefit at conservative targets	Not separately graded	Low — TREAT found no mortality benefit even at the conservative target in diabetic CKD

Query 2 treats the cross-trial inconsistency (TREAT showing excess stroke risk; CHOIR and CREATE not showing it) as the dominant factor and downgrades certainty to Moderate. Query 3 treats a dose-response relationship — higher ESA dose associated with harm independent of Hb level achieved — as an upgrading factor and rates the same harm signal as High. Both

moves are individually defensible under GRADE methodology, but they are not the same move, and produce different headline certainty levels for the single most clinically important finding in this evidence base.

This divergence only became visible by running the same question twice under different scaffolding and comparing the outputs directly — it would not have been caught from either arm in isolation.

6. Ratings by Factor

Each arm rated 1–10 on factors relevant to evidentiary rigor and EBM protocol fidelity. Ratings reflect internal coherence and consistency with the known literature, not independent re-verification of every primary source.

Factor	Query 1 (Plain)	Query 2 (Module 45)	Query 3 (Module 45 + Shadow)
Factual accuracy	8	8.5	8
Numerical precision	7	7.5	6.5
EBM protocol fidelity	2	8	8.5
Surrogate vs. patient-important outcome discipline	5	8	9
Internal consistency	8	7.5	7
Source transparency / verifiability	4	7	7
Epistemic honesty / self-flagged uncertainty	3	8	9
Clinical usability for a Socratic case	8	7	7.5
Currency (most recent guideline/trial data)	9	5	6

7. Usability Comparison

Beyond content quality, the three arms differ substantially in how they would function inside an actual teaching or clinical workflow.

Dimension	Query 1 (Plain)	Query 2 (Module 45)	Query 3 (Module 45 + Shadow)
Time to useful output	Fastest — single pass	Slower — multi-step protocol made visible	Slowest — two voices to track, heaviest output
Cognitive load on reader	Low	Medium-high	High
Best curriculum	Cold-open / first	Mid-curriculum	Post-hoc critique step,

placement	exposure, before appraisal skills are taught	worked example, once GRADE/RoB2 vocabulary is known	after a learner has tried to find the consult's own weak points
Main failure mode	Confident tone with no visible appraisal trail — risks being read as more authoritative than it is	Self-contained but not self-correcting; nothing forces a second look at its own certainty calls	Introduces a second meta-analysis without reconciling it against the first — the Shadow is not immune to the problem it critiques

8. Recommended Sequencing for Teaching Use

Rather than treating one arm as categorically “best,” the three outputs are most useful in combination, sequenced to match what each is good at:

- Step 1 — Show the plain output (Query 1 style) cold. Have the learner predict the certainty/strength of evidence themselves before any structured appraisal is shown.
- Step 2 — Reveal the Module 45 structured run (Query 2 style). The learner checks their prediction against the explicit PICO, RoB2, and GRADE steps.
- Step 3 — Withhold the Shadow critique until Step 2 is complete. Ask the learner to find weaknesses in the Query 2 consult themselves before seeing the Shadow Inventory.
- Step 4 — Reveal the Query 2 vs. Query 3 GRADE divergence (Section 5 above) as the closing discussion point. This is the highest-yield teaching moment across all three arms: a documented case of identical protocol and largely identical evidence producing different certainty verdicts.

9. Bottom-Line Findings (All Arms)

- ESA therapy reliably corrects anaemia and reduces transfusion need in CKD patients — this is the most secure benefit across all three arms and all certainty grades applied (High).
- Pushing haemoglobin toward normal/high targets (>13 g/dL, or even >11–12 g/dL in some trials) does not improve survival, cardiovascular outcomes, or quality of life, and several major RCTs show harm (excess mortality, cardiovascular events, or stroke) at higher targets.
- Quality-of-life benefit is the least secure claim across all three arms, with Low-to-Moderate certainty and inconsistent findings across CHOIR, CREATE, and TREAT.
- Iron repletion (per PIVOTAL) is part of the benefit story, not a separate adjunct — ESA efficacy and safety are iron-status-dependent.
- None of the exact percentages or hazard ratios cited across any of the three arms have been verified against primary trial publications within this comparison; all three arms themselves flag this and recommend primary-source verification before any figure is used in a paper or teaching module.

10. Limitations of This Comparison

- This report compares the structure and internal consistency of three AI-generated outputs; it does not independently re-verify the underlying trial data against primary publications (CHOIR, CREATE, TREAT, NHCT, PIVOTAL, Cochrane 2023 NMA).
- The GRADE divergence identified in Section 5 may reflect a genuine methodological improvement in Query 3 (incorporating dose-response data), an artifact of which adversarial challenge happened to be applied at that step, or some combination of both — this comparison cannot adjudicate which, and primary-source review would be needed to resolve it.
- Ratings in Section 6 are qualitative judgements made for the purpose of this comparison, not a validated psychometric instrument.